

RESEARCH ON THE DEVELOPMENT OF AN EXPERIMENTAL MODEL FOR DISEASE DETECTION IN VEGETABLE PLANTS

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Abstract: Plant diseases represent a major threat to global food security, causing annual losses estimated at 220 billion dollars worldwide. Existing mobile-based diagnostic solutions show reduced reliability in real field conditions due to uncontrolled lighting and variable image acquisition distances. This paper presents the design and development of an experimental system for automated disease detection in vegetable plants, integrating a controlled hardware acquisition station with an artificial intelligence algorithm based on YOLOv8. The hardware system includes an ESP32 microcontroller, an ultrasonic distance sensor, a light intensity sensor, and an automatic LED illumination ring, all coordinated through the MQTT protocol. The detection model was trained on a dedicated dataset and validated through performance metrics. Results demonstrate the system's capability to identify plant diseases under standardized conditions, reducing the laboratory-to-field performance gap.

KEY WORDS: YOLOv8, detection, control, capture, plants

1. Introduction

Plant diseases represent one of the main causes of agricultural losses, with recent estimates indicating annual losses of approximately 220 billion dollars worldwide [1]. In vegetable crops, late detection can compromise the entire production within a short period of time.

Existing software solutions based on mobile applications present limitations under real conditions. Their performance drops from 95% in the laboratory to 70-85% in the field, due to lighting variations and uncontrolled distance between the camera and the leaf [2].

This paper presents an integrated experimental model that combines a hardware station with a controlled environment and a YOLOv8 algorithm, reducing the influence of external factors on diagnostic accuracy.

2. Current State of the Art

The current market offers a variety of solutions for plant disease detection, differentiated by the technology used and the field of applicability.

One of the most widely used diagnostic platforms is Plantix [3]. The platform, with its application interface illustrated in Figure 1, operates as a virtual consultant accessible through a mobile device. The system identifies hundreds of pathogens based on images captured by the user, but its performance depends on image quality and lighting conditions.

RESEARCH ON THE DEVELOPMENT OF AN EXPERIMENTAL MODEL FOR DISEASE DETECTION IN VEGETABLE PLANTS



Fig. 1. User Interface of the Plantix Application [3]

Agrio [4] represents a more advanced approach, integrating artificial intelligence with parcel-level surveillance. It offers broader coverage, but high costs and the lack of adaptation to controlled environments such as greenhouses represent significant limitations.

In the academic environment, the FarmAid platform [5], illustrated in Figure 2, proposes a modular architecture with an integrated camera and sensors, designed for disease detection in greenhouses. The platform's performance is superior to mobile applications, but the high technical complexity and cost limit its accessibility.



Fig. 2. The FarmAid Platform [5]

Compared to the presented solutions, the experimental model proposed in this paper aims to combine a controlled acquisition environment with a high-performance detection algorithm, at a reduced cost and with accessible technical complexity.

3. System General Architecture

The disease detection system for vegetable plants is structured into several interdependent subsystems: the mechanical subsystem provides physical support in a closed enclosure that limits the influence of external light; the measurement subsystem extracts data from the analyzed environment; the control subsystem, based on the ESP32-C6 microcontroller; the actuation subsystem ensures uniform capture conditions and system status indication; the software subsystem consisting of the YOLOv8 model, the Python application and the database; the processing and analysis subsystem represented by the laptop; the communication subsystem, which transmits data through the MQTT protocol; and the power supply subsystem. Figure 3 presents the IDEF0 diagram of the system.

RESEARCH ON THE DEVELOPMENT OF AN EXPERIMENTAL MODEL FOR DISEASE DETECTION IN VEGETABLE PLANTS

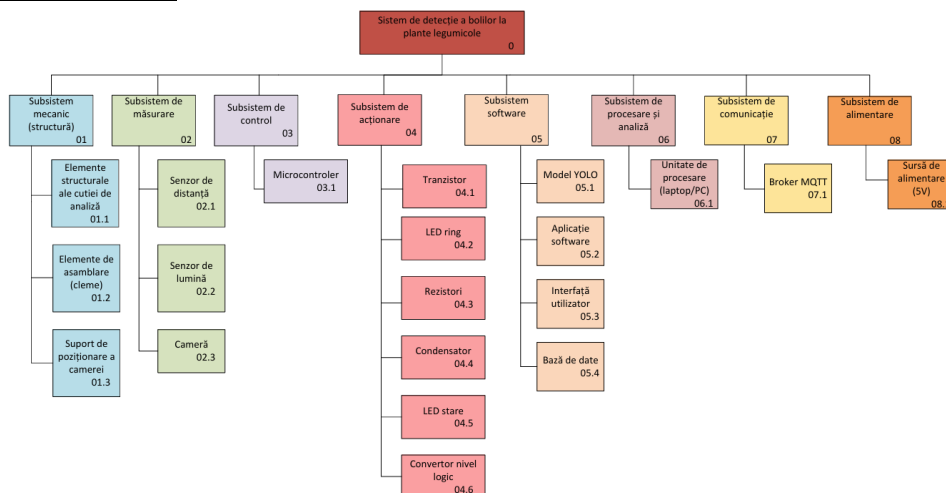


Fig. 3. The IDEF0 diagram of the system

To identify and rank the system functions, the FAST method (Function Analysis System Technique) was applied, with the diagram illustrated in Figure 4. The main function is image processing (F3), followed by diagnosis generation (F4) and leaf image capture (F1). The illumination assurance and external influence limitation functions support the acquisition process and contribute to obtaining consistent results.

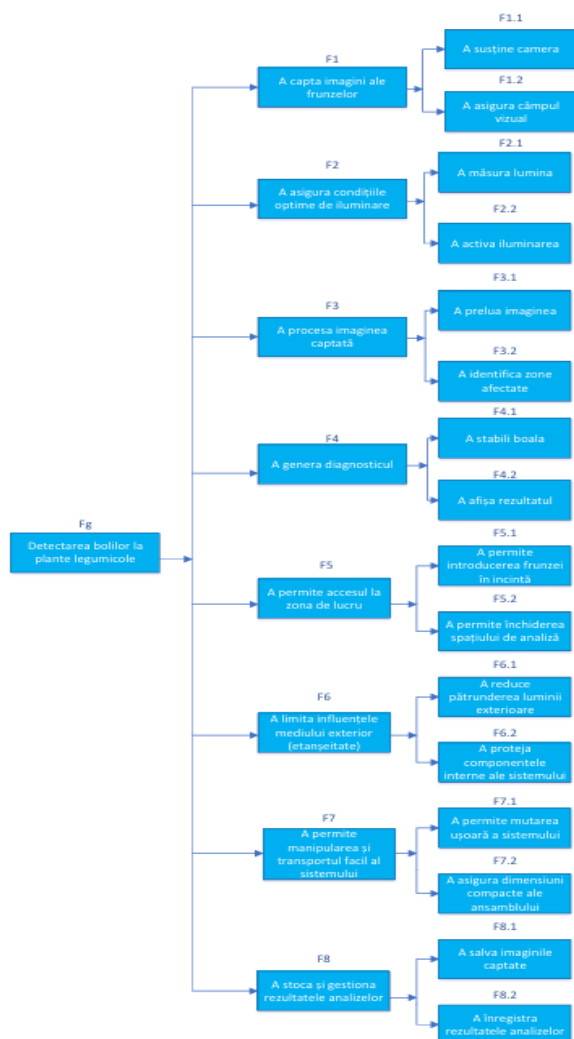


Fig. 4. The FAST diagram

RESEARCH ON THE DEVELOPMENT OF AN EXPERIMENTAL MODEL FOR DISEASE DETECTION IN VEGETABLE PLANTS

The mechanical structure was designed in CATIA V5 and manufactured by FDM 3D printing using PLA filament. The assembly takes the form of a closed box, consisting of a base, lateral walls, upper wall, front wall, and a protective enclosure for the electronic components. The front wall acts as a sliding door guided by channels integrated in the lateral walls. The camera module, HC-SR04 and BH1750 sensors, the LED ring, and the RGB status LED are mounted on the upper wall. Figure 5 presents the 3D model of the assembly.

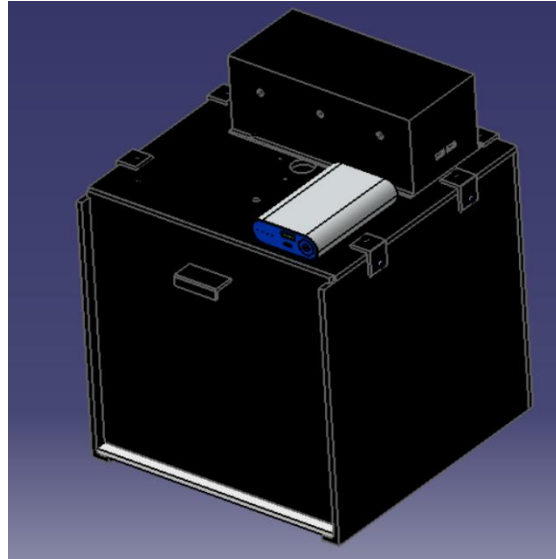


Fig. 5. The 3D model of the assembly

The system operates based on a logical sequence of operations, starting with initialization and ending with obtaining the result. The user starts the system, and the presence of the leaf is verified using the HC-SR04 sensor, whose reading is published to the MQTT broker and retrieved by the application. If no leaf is detected, the system remains in standby mode. After the leaf is detected, the BH1750 sensor measures the light intensity and, if it is below the set threshold, the microcontroller activates the LED ring through the MOSFET transistor. The user initiates the scan, and the camera captures the leaf image. This is processed by the YOLOv8 algorithm, and the results are displayed in the interface, saved in the SQLite database, and published to the MQTT broker. During the process, the RGB LED indicates the system status: blue for standby, red for scan in progress, and green for completed analysis.

4. Hardware Components

The electronic components were selected based on the functional specifications of the system and are presented in Table 1.

Table 1. Main components of the system

Component	Role	Technical Characteristics
ESP32-C6	Coordinates sensors and communication	RISC-V 32-bit, 160 MHz, WiFi 6, Bluetooth 5, 22 GPIO pins
Camera module OV3660	Leaf image capture	1920x1080 px, 64° angle, manual focus
LED ring	Uniform illumination of the analysis area	10W, 5V, 112 mm outer diameter
HC-SR04 sensor	Leaf presence detection	Range 2-450 cm, angle <15°,

RESEARCH ON THE DEVELOPMENT OF AN EXPERIMENTAL MODEL FOR DISEASE DETECTION IN VEGETABLE PLANTS

		Trigger/Echo interface
BH1750 sensor	Light intensity measurement	0-65535 lux, 1 lux resolution, I2C interface
MOSFET IRLZ44N	Illumination system control	N-channel, logic level, Vds 55V, Id 47A
Electrolytic Capacitor	Power supply stabilization	1000 μ F, 10V, radial mounting
10 k Ω resistor	Biassing (pull-down) in the control circuit	Metal film, 1/4W, $\pm$ 1% tolerance
220 Ω resistor	Current limiting	Metal film, 2W, $\pm$ 1% tolerance
RGB LED	System status indication	5 mm, 4 pins, common cathode
Logic level converter	HC-SR04 (5V) to ESP32 (3.3V) compatibility	Bidirectional, 4 channels, 5V/3.3V
Power supply (Powerbank)	System power supply	5000 mAh, 5V USB output

The electrical diagram of the system is presented in Figure 6. It illustrates the connections between the microcontroller, sensors, the MOSFET-controlled LED ring, the RGB LED, and the logic level converter.

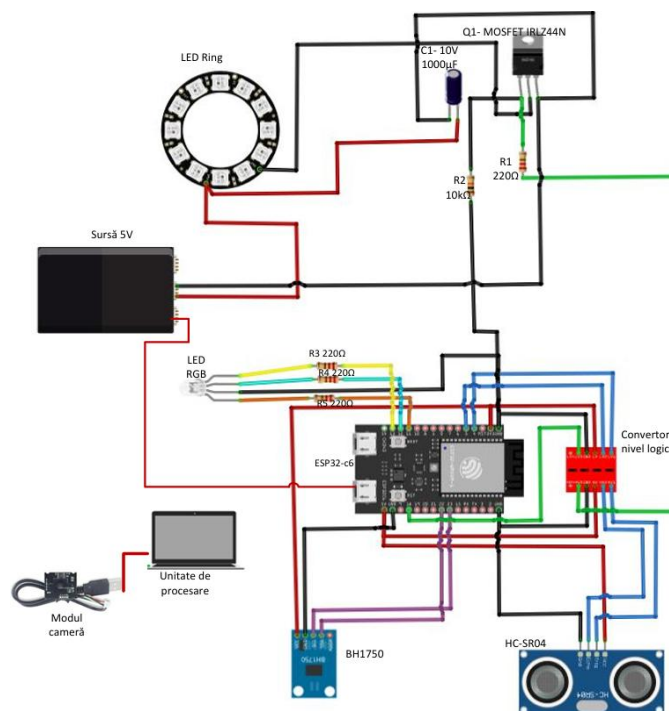


Fig. 6. The electrical diagram of the system

5. YOLOv8 Model Training

Model training was performed in Google Colab, using the YOLOv8n architecture, a variant optimized for low computational resources and short inference time. Four distinct models were trained, one for each crop: cucumbers [6], tomatoes [7], peppers [8], and eggplants [9], using datasets retrieved from the Roboflow platform. Table 2 presents the image distribution in each dataset, and Table 3 presents the training parameters.

RESEARCH ON THE DEVELOPMENT OF AN EXPERIMENTAL MODEL FOR DISEASE DETECTION IN VEGETABLE PLANTS

Table 2. The image distribution in each dataset

Plant	Total images	Training	Validation	Testing
Cucumbers	1368	1194	130	44
Tomatoes	9055	6462	1721	872
Peppers	1681	1177	336	168
Eggplants	600	420	120	60

Table 3. The training parameters

Parameter	Value
Architecture	YOLOv8n
Epochs	50
Image size	640 x 640 px
Batch size	16

The performance of each model was evaluated through validation metrics generated by YOLOv8, summarized in Table 4.

Table 4. Performance metrics per model

Model	mAP50	Maximum F1	Maximum detection rate
Cucumbers	0.830	0.84	0.94
Tomatoes	0.558	0.56	0.81
Peppers	0.852	0.83	0.92
Eggplants	0.995	0.99	1.00

The comparative performance of the four models is presented in Figure 7 through the mAP50, maximum F1, and maximum detection rate metrics. It can be observed that the eggplant model achieves the highest performance, with a mAP50 of 0.995, due to the high quality of the dataset and the visually distinct symptoms between classes. The model with the lowest performance is the one for tomatoes, explained by the large number of classes (10) and the visual similarity between them. The cucumber and pepper models achieve satisfactory results.

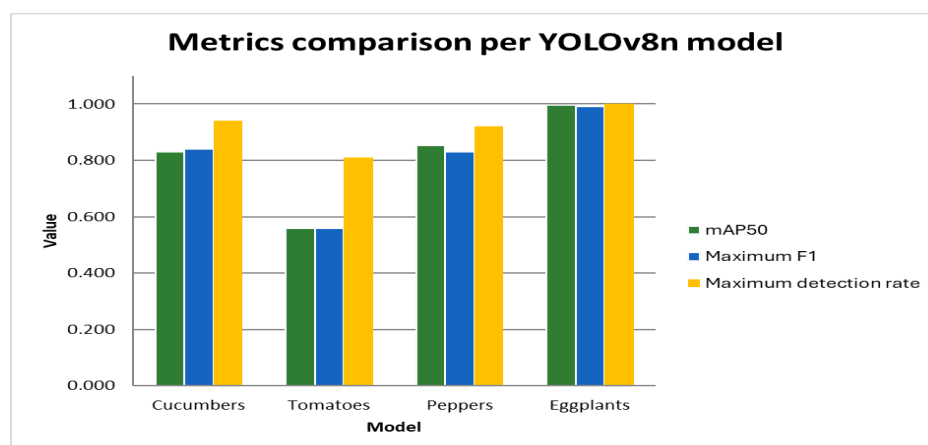


Fig. 7. Metrics comparison per YOLOv8n model

Figures 8-10 present examples of detections performed on images from the test set.

RESEARCH ON THE DEVELOPMENT OF AN EXPERIMENTAL MODEL FOR DISEASE DETECTION IN VEGETABLE PLANTS

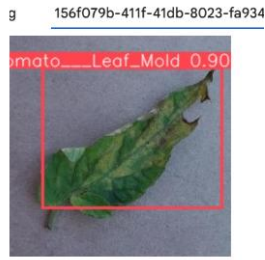


Fig. 8. Detection - tomatoes

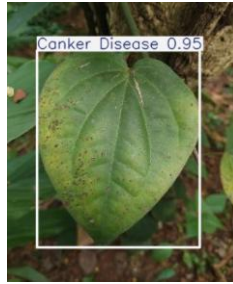


Fig. 9. Detection – peppers



Fig. 10. Detection - eggplants

6. Software Implementation

The software application was developed in Python, with a client-server architecture, where the web browser communicates with a Flask server through HTTP requests, and the server coordinates YOLOv8 inference, MQTT communication, and result storage. The four trained models are loaded once at server startup to eliminate latency at each scan.

Figure 11 presents the application architecture.

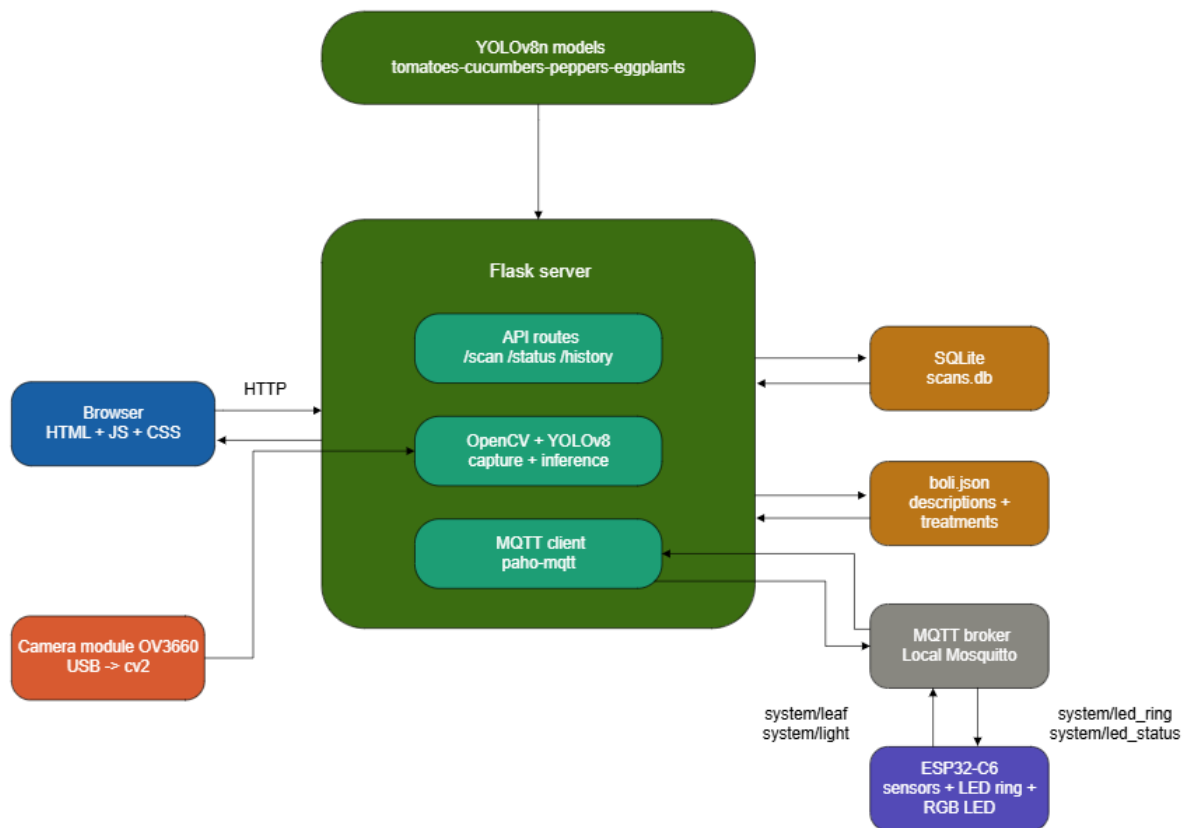


Fig. 11. The architecture of the software application

The web interface, illustrated in Figure 12, is organized into four areas: the navigation bar with a dropdown for plant selection and the “Start system” and “Start scan” buttons; the status panel with automatically updated indicators for leaf presence, illumination level and time of last scan; the visualization area with a live video stream and the last processed capture; and the history table with the last 20 scans.

RESEARCH ON THE DEVELOPMENT OF AN EXPERIMENTAL MODEL FOR DISEASE DETECTION IN VEGETABLE PLANTS

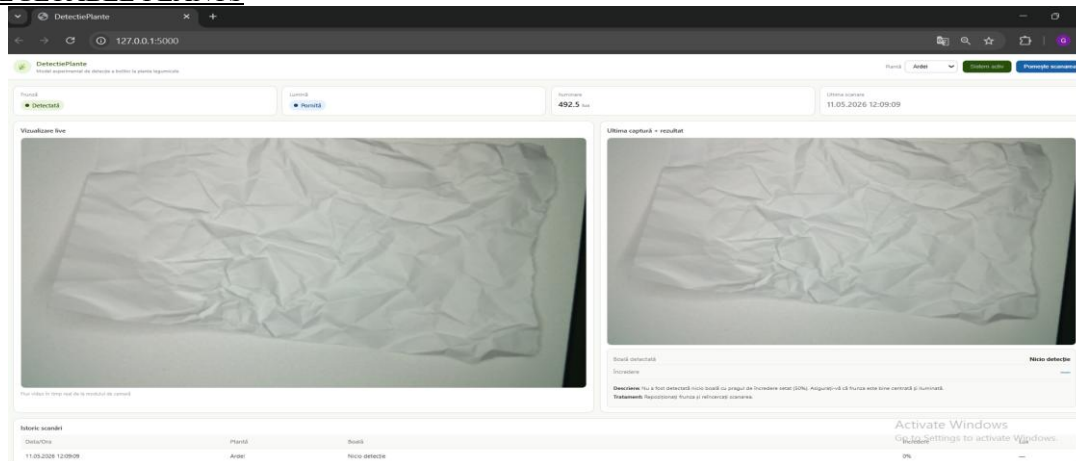


Fig. 12. Web Interface during a functionality test session

Communication between the application and hardware is achieved through the MQTT protocol, using the Mosquito broker running locally. Four topics are defined through which the ESP32 publishes leaf presence and illumination level in lux, while the application controls the RGB LED status and the LED ring. The illumination threshold was set experimentally at 500 lux to ensure stable activation of the LED ring during scanning. Scan results are automatically stored in the SQLite database, regardless of whether a disease was detected or not, including the plant type, identified disease, confidence score, date and time, and the path to the captured image.

7. Conclusions

This paper presents an integrated experimental system for disease detection in vegetable plants, combining a controlled-environment hardware station with YOLOv8n algorithms. Illumination control and leaf presence detection ensure uniform acquisition conditions, reducing the influence of external factors. Compared to existing mobile solutions, the system provides an accessible environment, representing a viable alternative for greenhouse use. Future developments aim to extend the number of crops and improve datasets. and integrate cloud connectivity.

8. References

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